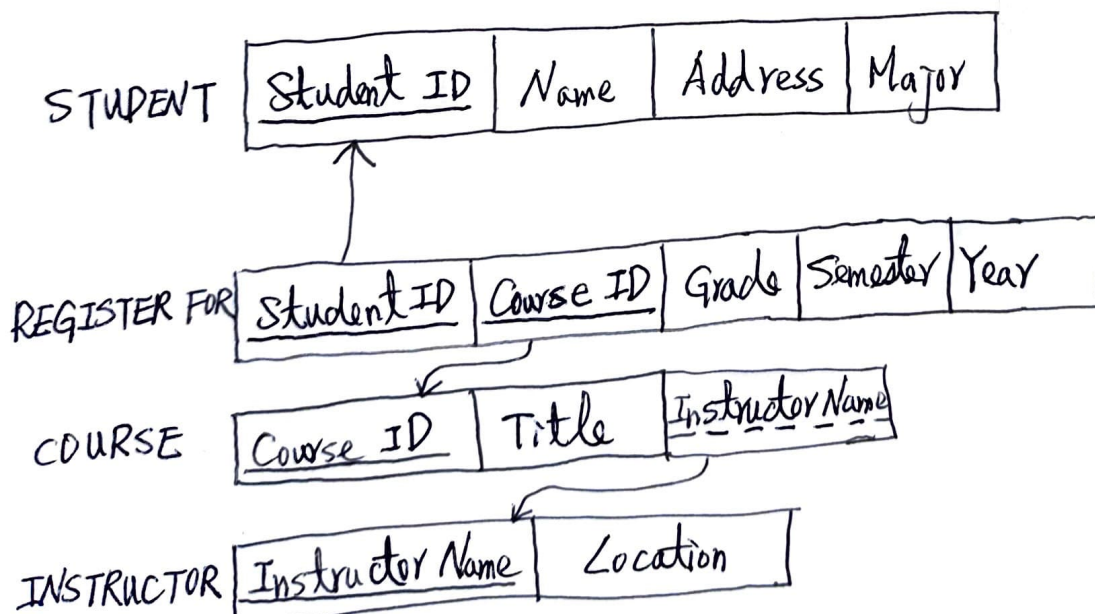
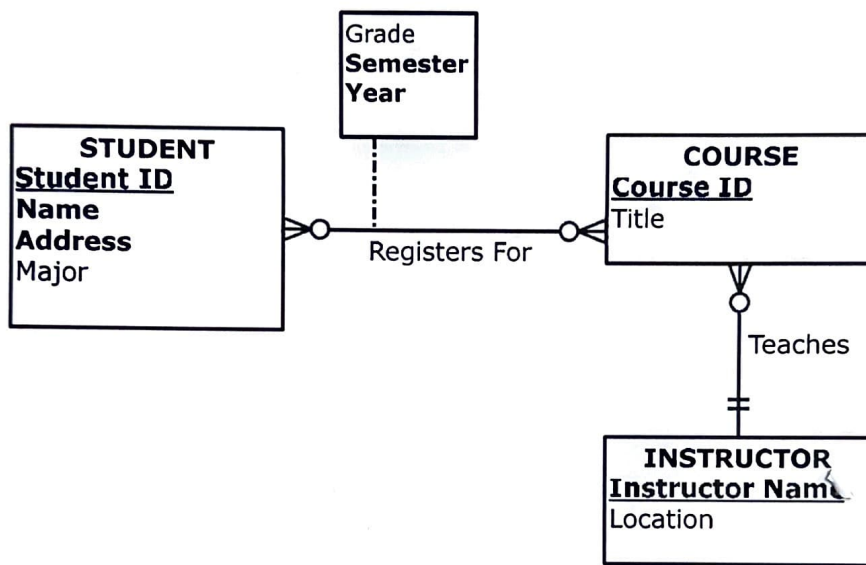


Homework #2

Due: 11/06/2024 (Wednesday) 23:59

(Please turn in to 作業管理 of i學園Plus)

1. Please transform the following ER-diagram into a relational schema.



2.

The following table is a sample report of a company.

Project NUM.	Project Name	Employee Number	Employee Name	Job Class	Charge /Hour	Hours Billed	Total Charge
1	Mercury	001	John Chen	Elect. Eng.	650	13	8450
		002	David Lee	Comm. Tech.	600	16	9600
		004	Jane Kao	Comm. Tech	600	19	11400
2	Satellite	001	John Chen	Elect. Eng.	650	15	9750
		003	Peter Liu	Biol. Eng.	550	17	9350
3	Mars	004	Jane Kao	Comm. Tech.	600	18	10800
		002	David Lee	Comm. Tech.	600	14	8400

If the following relational schema is the base relation of the above table, its name is **ProjectEmp**. If (P_NO, E_NO) is the Primary key of this relation.

ProjectEmp

<u>P_NO</u>	<u>E_NO</u>	P_Name	E_Name	Job_Class	CHG_Hour	Hours
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The functional dependencies include:

- (1) $P_NO \rightarrow P_Name$,
- (2) $E_NO \rightarrow E_Name$,
- (3) $Job_Class \rightarrow CHG_Hour$,
- (4) $(P_NO, E_NO) \rightarrow Hours$

Please answer the following two questions.

- (a) Is the above table an 1NF relation? If not, please re-write the above table into an 1NF relation. *please re-write by yourself.*
- (b) Describe and illustrate the process of normalizing the above relation to 3NF relations. (You have to normalize it to 2NF relations first, then to 3NF relations.)

(b) ①

<u>P-NO</u>	P-Name
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(2NF, 3NF)

<u>E-NO</u>	E-Name
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(2NF, 3NF)

<u>P-NO</u>	<u>E-NO</u>	Job-class	CHG-Hour	Hours
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(2NF, but not 3NF)

②

<u>Job-Class</u>	CHG-Hour
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(3NF)

<u>P-NO</u>	<u>E-NO</u>	Job-Class	Hours
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(3NF)